

Module

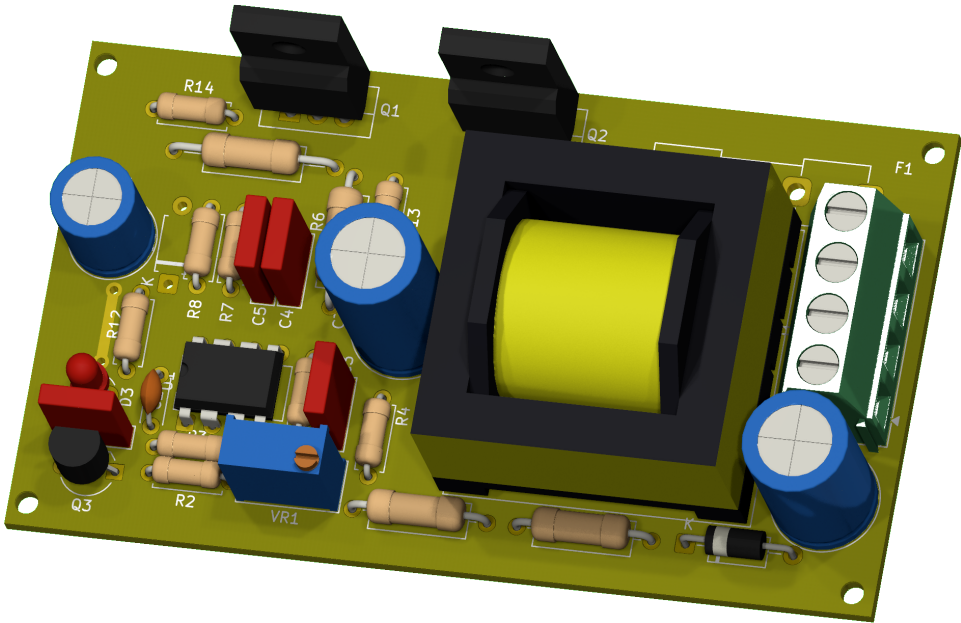
Variant: RELEASED

Rev 1.1.0

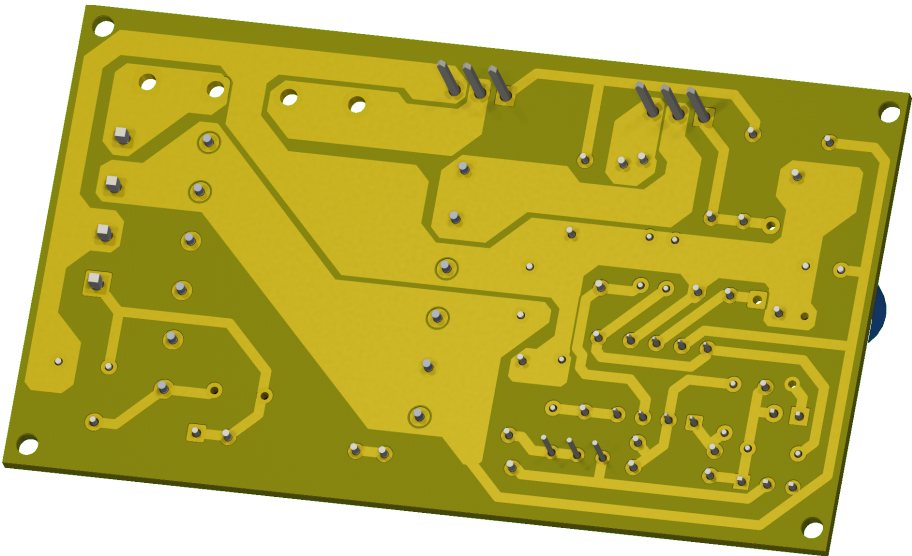
2026-01-16

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TOP VIEW



BOTTOM VIEW



DRAFT - Early concept stage, schematic drafted.

PRELIMINARY - Minor schematic changes are expected, ready for layout planning.

CHECKED - DRC completed, schematic and layout ready for review.

RELEASED - Design ready for manufacturing.

NOTE STYLING CONVENTION

DESIGN NOTE:
Example text for
informational
design notes.

DESIGN NOTE:
Example text for
debug notes.

DESIGN NOTE:
Example text for
cautionary
design notes.

DESIGN NOTE:
Example text for
critical design
notes.

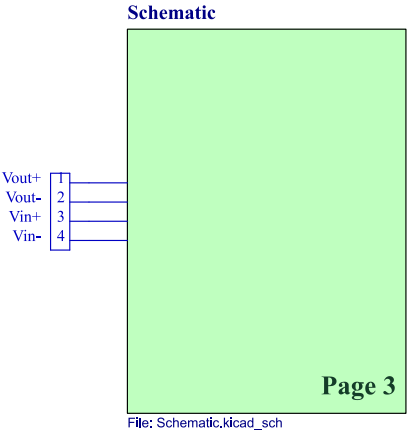
LAYOUT NOTE:
Example text for
critical layout
guidelines.

Date: 16-Jan-2026

	Comments:	Company: FR		Variant: RELEASED	Git Hash: 8a80b75
		Board Name: Module		Project Name: Step-up module 12:450 V	
	Sheet Title:	File Name: StepUp_module_12to450V.kicad_sch	Designer: FR	Date: 2025-12-20	Revision: 1.1.0
	Sheet Path: /		Reviewer: NA	Size: A3	Sheet: 1 of 4

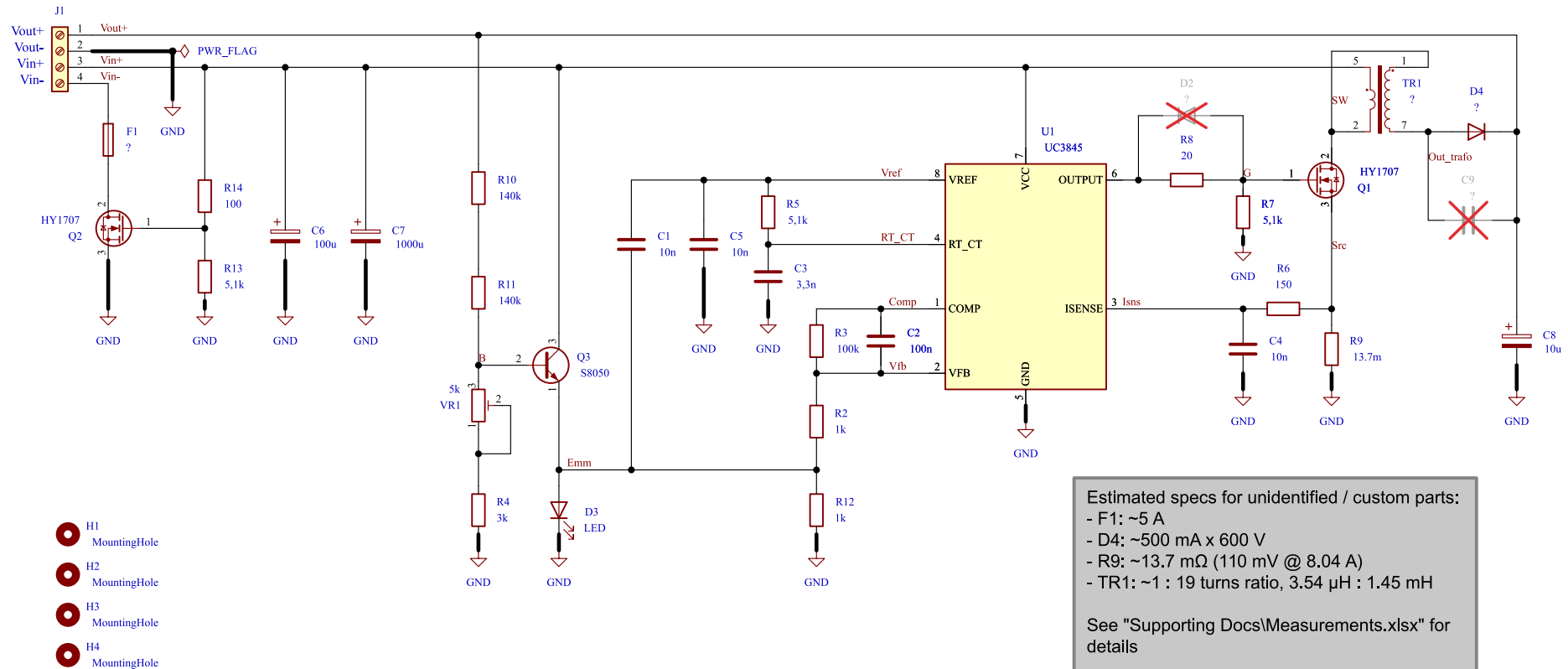
[2] Block Diagram

MODULE INTERFACE:
Vin: 9 V to 24 V according
to AliExpress product page
Vout: 200 V to 450 V,
adjustable through VR1



	Comments: AliExpress module example (link)		Company: FR		Variant: RELEASED		Git Hash: 8a80b75	
			Board Name: Module			Project Name: Step-up module 12:450 V		
	Sheet Title: Block Diagram		File Name: Block Diagram.kicad_sch		Designer: FR		Date: 2025-12-20	
							Revision: 1.1.0	
		Sheet Path: /Block Diagram/			Reviewer: NA		Size: A4	
							Sheet: 2 of 4	

[3] Schematic



Comments:

Company:

Variant:

Git Hash:

FR

RELEASED

8a80b75

Board Name:

Project Name:

Module

Step-up module 12:450 V

Sheet Title:

File Name:

Designer:

Date:

Revision:

Schematic

Schematic.kicad_sch

FR

2025-12-20

1.1.0

Sheet Path:

Reviewer:

Size:

Sheet:

/Block Diagram/Schematic/

NA

A4

3 of 4

[4] Revision History

Version 1.0.0 - 2025-12-19

First issue

Schematic capture complete, except for parts:

- not populated (D2, C9);
- with unreadable or absent markings (F1, D4, R9, TR1).

CI failed on release - do not use.

Version 1.0.1 - 2025-12-21

Changed

Changelog updates only - issue with missing changelog on revision 1.0.0

Version 1.1.0 - 2026-01-16

Changed

Replaced UC2843A by UC3845 - output signal is active only once every two cycles, indicating the IC matches UCx845 functionality.

Added

Measurement data (waveforms, transformer characteristics). See "Supporting docs"

	Comments:	Company: FR		Variant: RELEASED	Git Hash: 8a80b75
		Board Name: Module		Project Name: Step-up module 12:450 V	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: FR	Date: 2025-12-20	Revision: 1.1.0
	Sheet Path: /Revision History/		Reviewer: NA	Size: A4	Sheet: 4 of 4